RESEARCH METHODS & PROFESSIONAL PRACTICE

Modules & Course Objectives

COURSE OBJECTIVES

- **Professional Development & Upskilling:** Build on students' existing skills to create a strong foundation to develop their capstone idea.

- **Ideation & Iteration:** Create a clear starting point and narrow down and iterate to create a clear and strong Capstone idea.

- **Collaboration & Feedback:** Work collaboratively and engage in an iterative feedback and design loop with peers.

- **Project Development:** Iterate through the stages of project development over the course of the semester including: ideation, research, and prototyping.

- **Pitching & Presentation Skills:** Improve presentation and communication skills through individual and group presentations in front of peers, faculty, and industry professionals.

- **Craft a Clear Proposal:** Students will gather and summarize the key materials they've created throughout the semester to develop a clear and strong final Capstone Proposal.

All work in the course is to guide students in the creation and completion of a Capstone Proposal and Final Pitch Presentation for a project that improves the world or adds value to an experience, builds on existing precedents, and is technically feasible and properly scoped.

RUBRIC

When reviewing and approving Capstone project proposals, ATLAS faculty will consider how well the project demonstrates: Research, Feasibility, Preparedness and Relevance to CTD.

([Final Rubric for Proposal & Pitch])

IN-CLASS TIME STRUCTURE

With the exception of Day 01, each class period should have a maximum of 5-10 minutes of lecturing (often no lecture), with the remainder of each class period dedicated to work time / opportunities for students to talk with one another / opportunities for one-on-one check-ins with students.

COURSE STRUCTURE

Module 01: Introductions & Starting Point
Course Warm-Up & Establishing Your Starting Point (Weeks 1–3)
Foundations, Defining Personal Goals and Priorities, Building Community, Collaboration, and Presentation + Networking Skills

Introduce students to the course and each other through a Research and Design Sprint. Additionally, students will explore and document their personal motivations, interests, and skills and use that as a basis for capstone project brainstorming.

Deliverables: Time Allocation, Capstone Goals and Priorities, R&D Sprint Exercise, Intro Presentations, and Starting Point Assignment (Venn Diagrams/50 Ideas).

Module 02: Narrowing
From Starting Point to First Pitch (Weeks 3–6)
Ideation & Brainstorming, Creative Inspiration, Precedents Research, and Networking

Students will narrow down their initial starting points into a concrete idea through precedent research and synthesis.

Deliverables: Weekly Precedents Logs (*starting templates assigned each week*), Draft Pitch (*due beginning of Module 03*)

Module 03: Idea and Group Commitment (Weeks 6–9)

Networking, Adjusting, Integrating Feedback, Pitching, and Set-Up for Module 04

Students will integrate feedback from draft pitch presentations and commit to a final project idea as an individual or group. As this happens they will prepare for external expert conversations and establish a solid foundation to begin prototyping in Module 04.

Deliverables: Final Group and Idea Commitment at the End of Week 8

Module 04: Experimentation and Prototyping (Weeks 9-11)

External Expert Feedback, Testing Feasibility, Prototyping, Iteration, and Remaining Proposal Pieces

Each student will conduct experiments that will connect their idea to the technologies and methods needed to iterate on and complete their Capstone. The primary objective of these experiments is to determine what technologies and designs will help successfully bring their projects to life (or not), and will propel the students forward as they build their Capstone Projects. Additionally, throughout this module, students will create/prepare the remaining pieces of their proposal and presentation.

Deliverables: 3-6 prototype experiments per student (including 1-3 technical prototypes/experiments) with visual documentation, key points and findings from expert conversations and experiments (will be included in the final pitch/proposal). Remaining pieces of proposal reviewed will be included in Final Proposal Submission.

Module 05: Final Stretch (Weeks 12–14)

Final Presentation Rehearsal, Feedback Integration, Final Project Proposals, and Final Pitch Presentation

Students will polish up their final pitch decks, submit a completed proposal document and present and pitch their project to a panel. Panels should include at least one faculty member and may also include friends of ATLAS / external experts. The panel will determine whether projects have been approved to move into the Spring “Capstone Projects” course as well as the final grade on this portion of the class.

Deliverables: Final Presentation and Proposal